
Level 5

THE PICKY ARM

```
FAB-42 > pick-up arm ... INSTALLED
FAB-42 > pattern ....... LOADED
```

FAB-42 now has a pick-up arm. It watches the board, and the moment a specific pattern is complete, it lifts it off and stores it.

The arm is picky: it only takes the pattern if it is in exactly the given position and no other matter is touching it. Otherwise, it waits.

Count how many patterns each program really produces.

The nanofabricator was made to produce patterns.

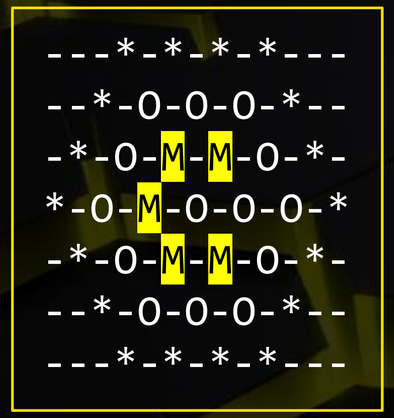
- You are given a list of programs and patterns
- A pattern is a group of connected matter
- Whenever the pattern is detected, it is submitted automatically
  - Submitted means that all its matter is removed from the board
  - “Pattern Detection” is explained on the next page
- **Simulate each program and count how many patterns were submitted**



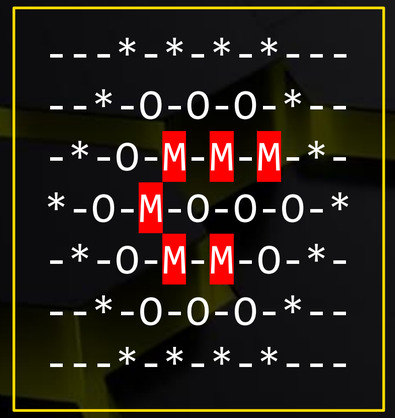
Pattern Detection

A pattern is submitted if:

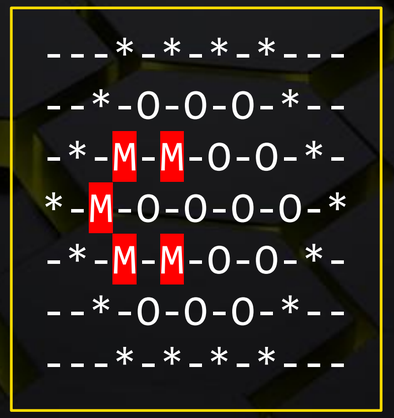
- All matter of the pattern is present on the board in the exact same position as it is given
- There may be other matter on the board, but it must not be adjacent to any matter of the pattern



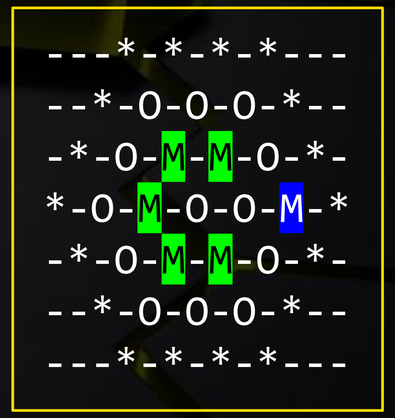
Example pattern



Invalid because of adjacent matter



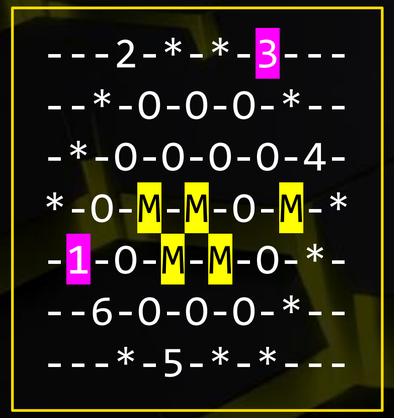
Invalid because it is shifted



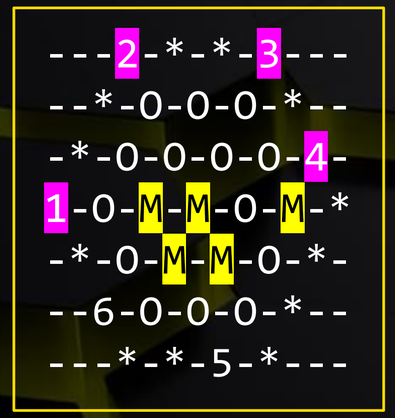
Valid

A tick now works in the following order:

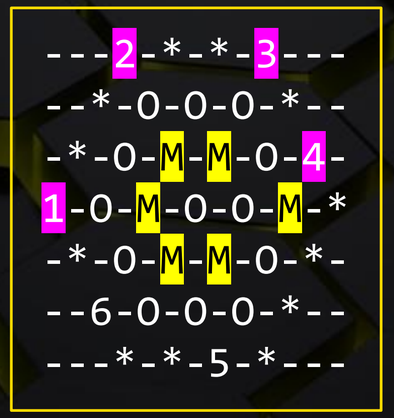
1. One instruction line is executed
2. Matter is created and destroyed
3. If a valid pattern is present, it is submitted



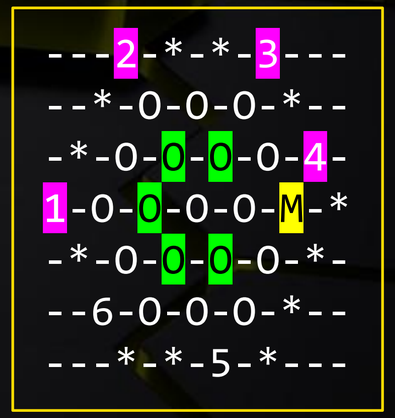
Example state



F T - T B -
is executed



Matter is
created and
destroyed



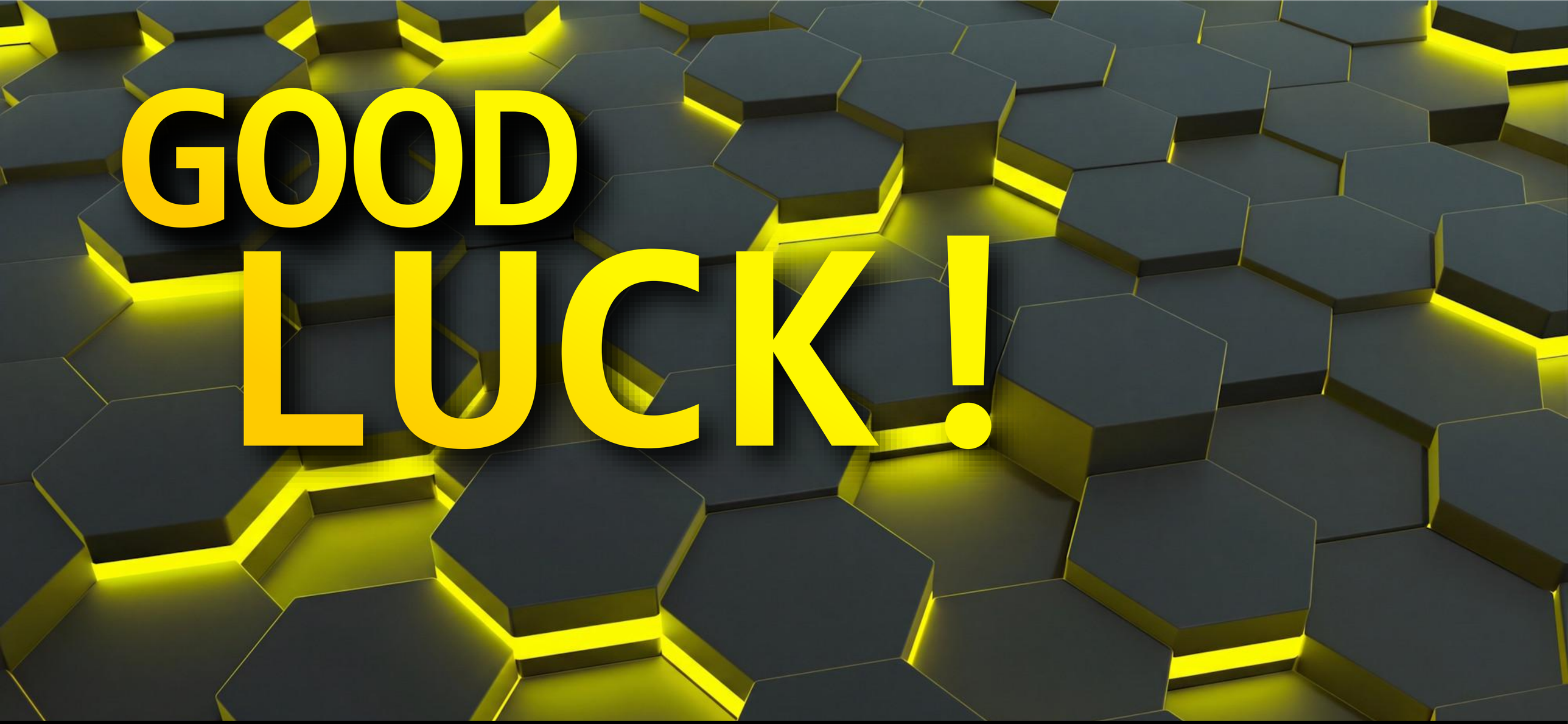
Pattern is
submitted

Input

Name		Description	Example
N		Number of programs	2 3 7 13 ---*-*-*-*--- --*-O-O-O-*-- -*-O-M-M-O-*- *-O-M-O-O-O-* -*-O-M-M-O-*- --*-O-O-O-*-- ---*-*-*-*--- 7 F T T F - - T - - T - - T B - B T - T F - F T - T B - B T - T F - F T - T B - B T - 3 7 13 ---*-*-*-*--- --*-O-O-O-*-- ...
repeated N times	Pattern dimensions	The **radius of the board**, the number of rows, and the number of columns, separated by spaces	
	Pattern	**The character representation of a pattern**	
	L	Number of instruction lines	
	Program	A list of L instruction lines Each line contains 6 instructions, separated by spaces	
	Empty line between programs		

Output

Name	Description	Example
Count (repeated N times)	The number of submitted patterns	3 5



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